Chapter 15. Multicollinearity and Biased Estimation

Multicollinearity, VIF, ridge regression, principal component regression

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The least squares estimator $\hat\beta = (X'X)^{-1}X'\mathbf{y}$ is the BLUE with covariance $\sigma^2(X'X)^{-1}$. When $X'X$ is nearly singular this covariance is huge, and a slightly biased estimator with much smaller variance can be far more accurate. This chapter studies the problem and two remedies.

Throughout we standardize: y is centred and each regressor is centred and scaled so that $\sum_i x_{ij} = 0$ and $\sum_i x_{ij}^2 = 1$. Then X is $n \times p$ (no intercept column), $X'X$ is the correlation matrix of the regressors, and $X'\mathbf{y}$ is proportional to their correlations with y. Coefficients on the original scale are recovered from $\tilde\beta_j = \hat\beta_j\, s_y/s_j$ and $\tilde\beta_0 = \bar y - \sum_j \tilde\beta_j \bar x_j$, where s_j is the scale used for x_j.

15.1 Multicollinearity

Multicollinearity means that the regressors are (nearly) linearly dependent. Exact dependence makes X rank-deficient and $\hat\beta$ undefined; near dependence makes $(X'X)^{-1}$ huge and $\hat\beta$ unstable.

Example 15.1 (Two regressors)

With two standardized regressors with correlation r_{12},

$$X'X = \begin{pmatrix} 1 & r_{12} \\ r_{12} & 1 \end{pmatrix}, \qquad \operatorname{Var}\hat\beta = \frac{\sigma^2}{1-r_{12}^2}\begin{pmatrix} 1 & -r_{12} \\ -r_{12} & 1 \end{pmatrix},$$

so $\operatorname{Var}\hat\beta_j = \sigma^2/(1-r_{12}^2) \to \infty$ as $|r_{12}| \to 1$, and $\operatorname{Corr}(\hat\beta_1, \hat\beta_2) = -r_{12}$. With $r_{12} = 0.99$ the variances are 50 times those under orthogonal regressors, and confidence intervals $\hat\beta_j \pm t_{\alpha/2}(n-3)\sqrt{MSE/(1-r_{12}^2)}$ are 7 times as wide.

Eigenvalues. If $\lambda_1, \dots, \lambda_p$ are the eigenvalues of $X'X$,

$$\sum_j \operatorname{Var}\hat\beta_j = \sigma^2 \operatorname{tr}(X'X)^{-1} = \sigma^2 \sum_j \frac{1}{\lambda_j},$$

so a single small eigenvalue blows up the total variance. The eigenvector of a small eigenvalue shows which combination of regressors is nearly constant.

Theorem 15.1 (Variance inflation factor)

Let R_j^2 be the coefficient of determination of the regression of x_j on the other regressors. Then c_{jj}, the j-th diagonal entry of $(X'X)^{-1}$, equals $(1-R_j^2)^{-1}$, so

$$\operatorname{Var}\hat\beta_j = \frac{\sigma^2}{1-R_j^2}.$$

Proof. Write $X = (\mathbf{x}_j, X_{-j})$. By the partitioned-inverse formula,

$$c_{jj} = \{\mathbf{x}_j'\mathbf{x}_j - \mathbf{x}_j'X_{-j}(X_{-j}'X_{-j})^{-1}X_{-j}'\mathbf{x}_j\}^{-1} = \{\mathbf{x}_j'(I - H_{-j})\mathbf{x}_j\}^{-1},$$

which is the inverse of the residual sum of squares of $\mathbf{x}_j$ on X_{-j}. Since $\mathbf{x}_j$ is centred with $\mathbf{x}_j'\mathbf{x}_j = 1$, its total sum of squares is 1, so the residual sum of squares is $1 - R_j^2$. ∎

$\mathrm{VIF}_j = (1-R_j^2)^{-1}$ is the **variance inflation factor**: the factor by which $\operatorname{Var}\hat\beta_j$ exceeds its value under orthogonal regressors. $\mathrm{VIF}_j > 10$ (i.e. $R_j^2 > 0.9$) is a common warning sign.

15.2 Ridge regression

Motivation. Under multicollinearity least squares remains unbiased but its variance explodes. Ridge regression adds kI to $X^\top X$; this biases the estimates but, for suitable k, always reduces mean squared error.

Measure the accuracy of an estimator b by the total mean squared error $E\|b-\beta\|^2$. For least squares,

$$E\|\hat\beta-\beta\|^2 = \operatorname{tr}\operatorname{Var}\hat\beta = \sigma^2\sum_j \frac{1}{\lambda_j}.$$

Definition (Ridge estimator)

For $k>0$, $\hat\beta(k)=(X'X+kI_p)^{-1}X'\mathbf{y}$, the minimizer of $\|\mathbf{y}-X\beta\|^2+k\|\beta\|^2$.

Adding k to the diagonal lifts all eigenvalues by k, so $X'X+kI$ is well conditioned. If $X'X=I$, then $\hat\beta(k)=\hat\beta/(1+k)$: ridge **shrinks** least squares towards $\mathbf{0}$. In general $\hat\beta(k)=(X'X+kI)^{-1}X'X\hat\beta$, which is biased: $E\hat\beta(k)=(X'X+kI)^{-1}X'X\beta\neq\beta$.

Canonical form. Let $X'X=P\Lambda P'$ with P orthogonal and $\Lambda=\operatorname{diag}(\lambda_1,\dots,\lambda_p)$. With $Z=XP$ and $\alpha=P'\beta$, the model is $\mathbf{y}=Z\alpha+\epsilon$ with $Z'Z=\Lambda$. Then $\hat\alpha=\Lambda^{-1}Z'\mathbf{y}$, $\operatorname{Var}\hat\alpha=\sigma^2\Lambda^{-1}$, and

$$\hat\alpha(k)=P'\hat\beta(k)=(\Lambda+kI)^{-1}Z'\mathbf{y}, \qquad \hat\alpha_j(k)=\frac{\lambda_j}{\lambda_j+k}\hat\alpha_j :$$

each canonical coefficient is shrunk, most strongly in directions with small λ_j. Since P is orthogonal, $\|\hat\beta(k)-\beta\|=\|\hat\alpha(k)-\alpha\|$.

Theorem 15.2 (Mean squared error of ridge)

$$E\|\hat\beta(k)-\beta\|^2=\sigma^2\sum_{j=1}^{p}\frac{\lambda_j}{(\lambda_j+k)^2}+k^2\sum_{j=1}^{p}\frac{\alpha_j^2}{(\lambda_j+k)^2}.$$

The first term (total variance) decreases in k and the second (squared bias) increases. There is always a $k>0$ with $E\|\hat\beta(k)-\beta\|^2<E\|\hat\beta-\beta\|^2$.

Proof [Supplement]. $\hat\alpha_j(k)$ has mean $\frac{\lambda_j}{\lambda_j+k}\alpha_j$ and variance $\frac{\lambda_j^2}{(\lambda_j+k)^2}\cdot\frac{\sigma^2}{\lambda_j}$; its bias is $-\frac{k}{\lambda_j+k}\alpha_j$. Adding variance and squared bias over j gives the formula. Call it $g(k)$. Then $g(0)=\sigma^2\sum 1/\lambda_j=E\|\hat\beta-\beta\|^2$, and

$$g'(k)=-2\sigma^2\sum_j\frac{\lambda_j}{(\lambda_j+k)^3}+2k\sum_j\frac{\lambda_j\alpha_j^2}{(\lambda_j+k)^3}.$$

At $k=0$, $g'(0)=-2\sigma^2\sum_j\lambda_j^{-2}<0$, so $g(k)<g(0)$ for small $k>0$. ■

In the orthonormal case $X'X=I$, $g(k)=\frac{p\sigma^2+k^2\|\alpha\|^2}{(1+k)^2}$, minimized at $k=p\sigma^2/\|\beta\|^2$ (set $g'(k)=0$). More generally, a separate constant k_j for each canonical direction, $\hat\alpha_j=\frac{\lambda_j}{\lambda_j+k_j}\hat\alpha_j$ (generalized ridge), has mean squared error $\sum_j\frac{\sigma^2\lambda_j+k_j^2\alpha_j^2}{(\lambda_j+k_j)^2}$, minimized at $k_j=\sigma^2/\alpha_j^2$.

Choosing k. The optimal k depends on unknown σ^2 and β. Practical choices:

- the **ridge trace**: plot $\hat\beta_j(k)$ against k and take the smallest k at which the coefficients stabilize;

- the plug-in rule $\hat{k} = p\hat{\sigma}^2/\hat{\beta}'\hat{\beta}$ (Hoerl, Kennard and Baldwin), possibly iterated with $\hat{\beta}(k)$ in place of $\hat{\beta}$;
- cross-validation, e.g. minimizing $\text{PRESS}(k) = \sum_i \{y_i - \mathbf{x}_i'\hat{\beta}_{(i)}(k)\}^2$. As in Theorem 10.3, with $A(k) = X(X'X + kI)^{-1}X' = (a_{ij})$ and $e_i = y_i - \mathbf{x}_i'\hat{\beta}(k)$,

$$\text{PRESS}(k) = \sum_{i=1}^{n} \Big(\frac{e_i}{1 - a_{ii}}\Big)^2,$$

so no refitting is needed (leave-one-out cross-validation).

15.3 Principal component regression

Let $\mathbf{e}_1, \dots, \mathbf{e}_p$ be orthonormal eigenvectors of $X'X$ with eigenvalues $l_1 \geq \cdots \geq l_p > 0$, and $E_r = (\mathbf{e}_1, \dots, \mathbf{e}_r)$. The **principal components** are the columns of $Z_r = XE_r$, and they are orthogonal: $Z_r'Z_r = \text{diag}(l_1, \dots, l_r)$.

All components ($r = p$). $X\beta = XE_pE_p'\beta = Z\alpha$ with $\alpha = E_p'\beta$, and

$$\hat{\beta} = E_p\hat{\alpha} = E_p(Z'Z)^{-1}Z'\mathbf{y} = \sum_{k=1}^{p} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k'X'\mathbf{y}, \qquad \text{Var}\,\hat{\beta} = \sigma^2 \sum_{k=1}^{p} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k',$$

which is just least squares.

Leading components ($r < p$). Regress $\mathbf{y}$ on Z_r only and transform back:

$$\tilde{\beta}_r = E_r(Z_r'Z_r)^{-1}Z_r'\mathbf{y} = \sum_{k=1}^{r} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k'X'\mathbf{y}.$$

Theorem 15.3 (Principal component regression)

$$E\tilde{\beta}_r = \beta - \sum_{k=r+1}^{p} \mathbf{e}_k\mathbf{e}_k'\beta, \qquad \text{Var}\,\tilde{\beta}_r = \sigma^2 \sum_{k=1}^{r} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k',$$

and

$$E\|\tilde{\beta}_r - \beta\|^2 = \sigma^2 \sum_{k=1}^{r} \frac{1}{l_k} + \sum_{k=r+1}^{p} (\mathbf{e}_k'\beta)^2.$$

Proof. Using $X'X = \sum_h l_h\mathbf{e}_h\mathbf{e}_h'$ and orthonormality, $E\tilde{\beta}_r = \sum_{k\leq r} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k'X'X\beta = \sum_{k\leq r} \mathbf{e}_k\mathbf{e}_k'\beta$, and $\beta = \sum_{k\leq p} \mathbf{e}_k\mathbf{e}_k'\beta$. Next,

$$\text{Var}\,\tilde{\beta}_r = \sigma^2 \sum_{k,k'\leq r} \frac{1}{l_k l_{k'}}\mathbf{e}_k\mathbf{e}_k'X'X\mathbf{e}_{k'}\mathbf{e}_{k'}' = \sigma^2 \sum_{k\leq r} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k'.$$

The trace of the variance plus the squared length of the bias gives the mean squared error. ■

Dropping the components with the smallest l_k removes the largest variance terms σ^2/l_k at the cost of the bias $\sum_{k>r}(\mathbf{e}_k'\beta)^2$. This pays off when the small-eigenvalue directions carry little of β. PCR is a "hard" version of ridge: ridge shrinks every canonical coefficient by $l_k/(l_k + k)$, PCR keeps the first r and sets the rest to 0.

Choosing r. Common rules: keep components until the cumulative proportion $\sum_{k\leq r} l_k / \sum_k l_k$ (the share of the total variance of the standardized regressors) is large enough; look for an elbow in the scree plot of l_k; or choose r by cross-validation. Note that the components are chosen without looking at y; partial least squares instead builds components that are correlated with y.